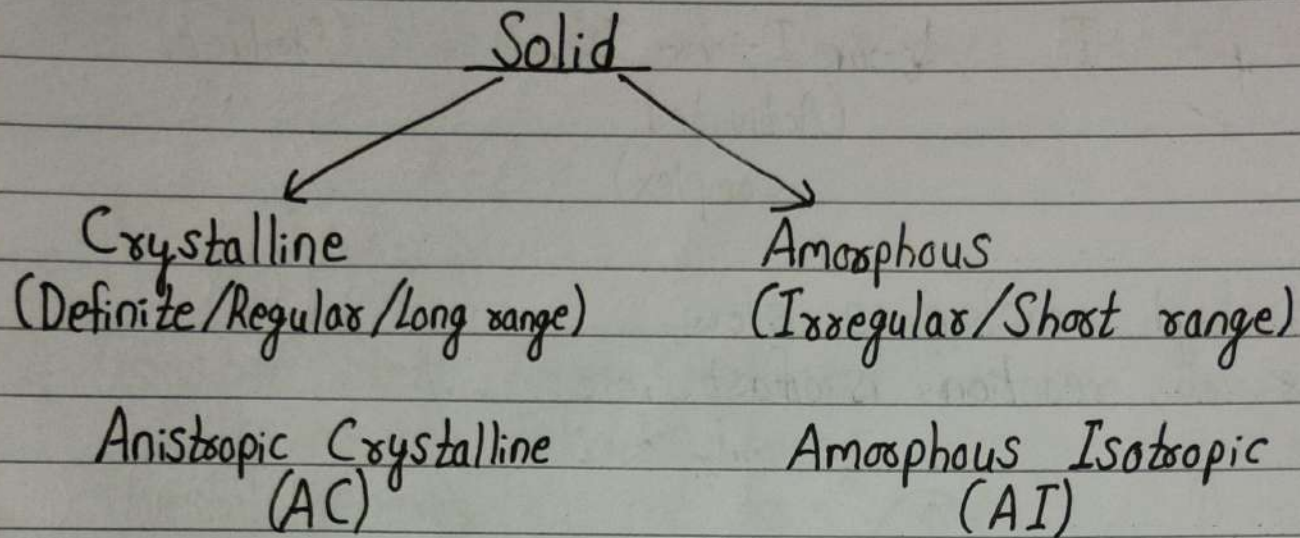
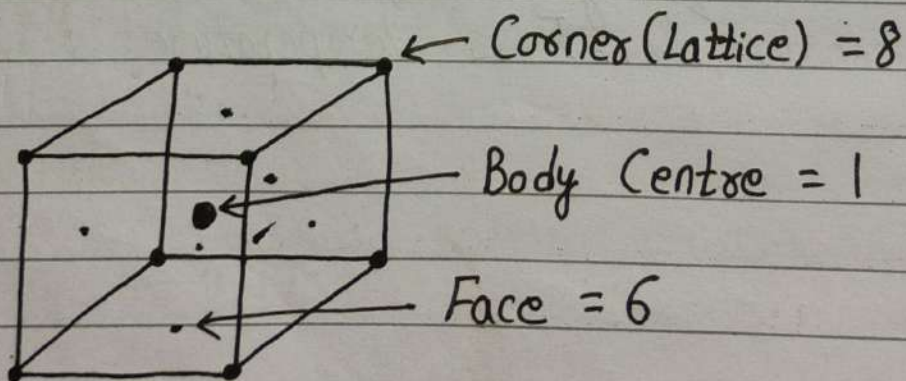
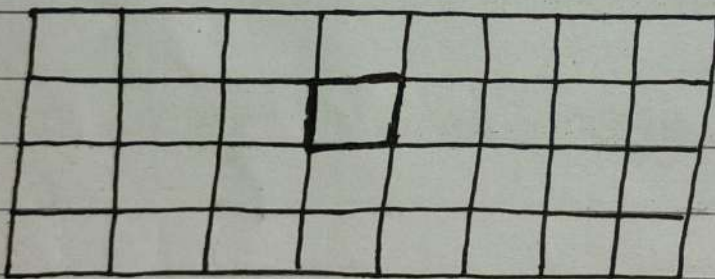


Module - I: Solid State

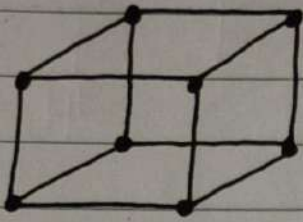


Unit Cell

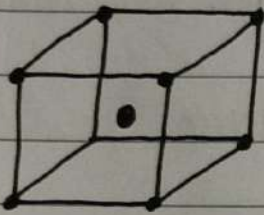
It is the smallest part of the crystal.



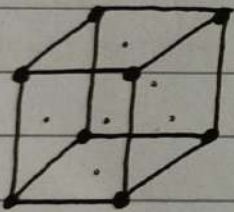
Types of Unit Cell (Cubic) (Barraise Lattice)



Simple Cubic (Primitive U.C.)

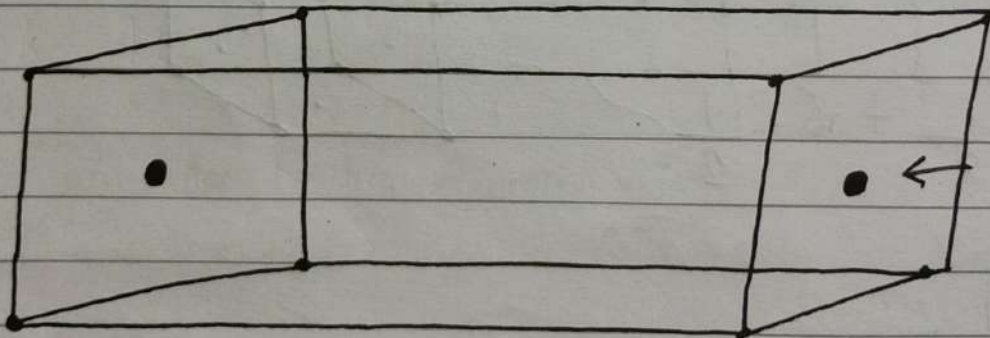


Body Centred Cubic (BCC)
[Corners + body centre]



Face Centred Cubic (FCC)
[Corners + face]

Cuboid

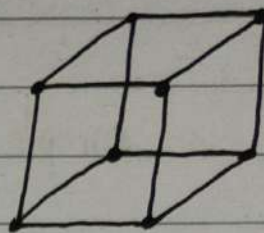


End Centred Cuboid (ECC)
[Corners + End Face]

Number of Atoms in Per Unit Cell (Z)

(i) Simple Cubic

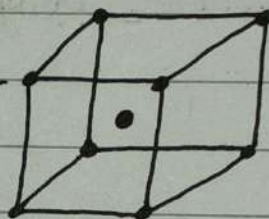
$$Z = 8 \times \frac{1}{8}$$



$$[Z = 1]$$

(ii) BCC

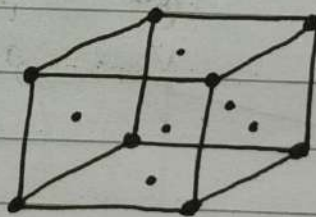
$$Z = 8 \times \frac{1}{8} + 1$$



$$[Z = 2]$$

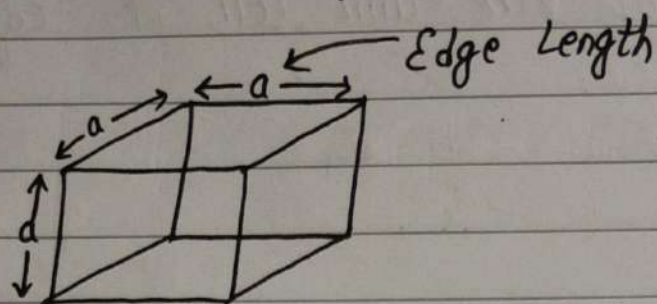
(iii) FCC

$$Z = 8 \times \frac{1}{8} + 6 \times \frac{1}{2}$$



$$[Z = 4]$$

Calculation of Density of Unit Cell



$$\text{Volume of U.C.} = lbh = a \cdot a \cdot a = a^3$$

$$\text{No. of atoms in per unit cell} = Z$$

$$\text{Atomic mass of element} = M$$

$$\text{Density of U.C.} = \frac{\text{Mass of U.C.}}{\text{Volume of U.C.}} \quad \text{--- (1)}$$

Now,

$$\text{Mass of U.C.} = \text{No. of atoms in per unit cell} \times$$

$$\text{Mass of an atom } \cancel{\text{per unit cell}} \\ = Z \times m$$

$$\text{Mass of an atom (m)} = \frac{\text{Atomic Mass (M)}}{N_A}$$

So,

$$\text{Mass of U.C.} = Z \times \frac{M}{N_A}$$

From eqⁿ. (1)

$$\boxed{\rho = \frac{Z \times M}{N_A \times a^3}} \quad [\text{Unit is gm/cm}^3.]$$

Q) An element having BCC geometry has atomic mass 50 u. Calculate the density of the unit cell, if edge length is 290 pico-metre.

Solⁿ:

$$\text{BCC, } Z = 2$$

$$M = 50 \text{ u}$$

$$a = 290 \text{ pm} = 290 \times 10^{-10} \text{ cm}$$

$$\therefore S = \frac{Z \times M}{N_A \times a^3} = \frac{2 \times 50}{6.022 \times 10^{23} \times (290 \times 10^{-10})^3} = \cancel{5.12 \times 10^{-15} \text{ gm/cm}^3}$$

$$\therefore S = 6.80 \text{ gm/cm}^3.$$

~~Q) An element having BCC geometry has~~

Q) The density of chromium metal is 7.2 gm/cm^3 . If the unit cell is cubic with edge length of 289 pm, determine the type of unit cell.
($C_r = 52$ & $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$)

Solⁿ:

$$\text{Given, } S = 7.2 \text{ gm/cm}^3$$

$$a = 289 \text{ pm}$$

$$M = 52$$

$$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$$

We know,

$$S = \frac{Z \times M}{N_A \times a^3}$$

$$\text{or, } Z = \frac{\rho \times N_A \times a^3}{M}$$

$$\therefore Z = \frac{7.2 \times 6.02 \times 10^{23} \times (289)^3}{52}$$

$$= 2.0119$$

i.e. The unit cell is ^{type} BCC.

Q) Crystalline solid has BCC structure. Calculate the unit cell edge length, if the density of crystal is 4.24 gm/cm^3 .
(Atomic mass = 213)

Solⁿ:

$$\text{Given, } \rho = 4.24 \text{ gm/cm}^3$$

$$M = 213$$

$$N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$$

$$Z = 2$$

$$a = ?$$

We know,

$$\rho = \frac{Z \times M}{N_A \times a^3}$$

$$\text{or, } a^3 = \frac{Z \times M}{N_A \times \rho}$$

$$\text{or, } a = \sqrt[3]{\frac{Z \times M}{N_A \times \rho}}$$

$$\therefore a = \sqrt[3]{\frac{2 \times 213}{6.022 \times 10^{23} \times 4.24}} \\ = 5.505 \times 10^{-8} \text{ cm.}$$

i.e. $a = 550 \text{ pm.}$

9) An element has a bcc structure with a cell edge of 288 pm . The density of the element is 7.2 g/cm^3 . How many atoms are present in 20 Find the atomic mass?

Solⁿ:

Given, $Z = 2$

$a = 288 \text{ pm}$

$S = 7.2 \text{ g/cm}^3$

$N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$

$M = ?$

We know,

$$S = \frac{Z \times M}{N_A \times a^3}$$

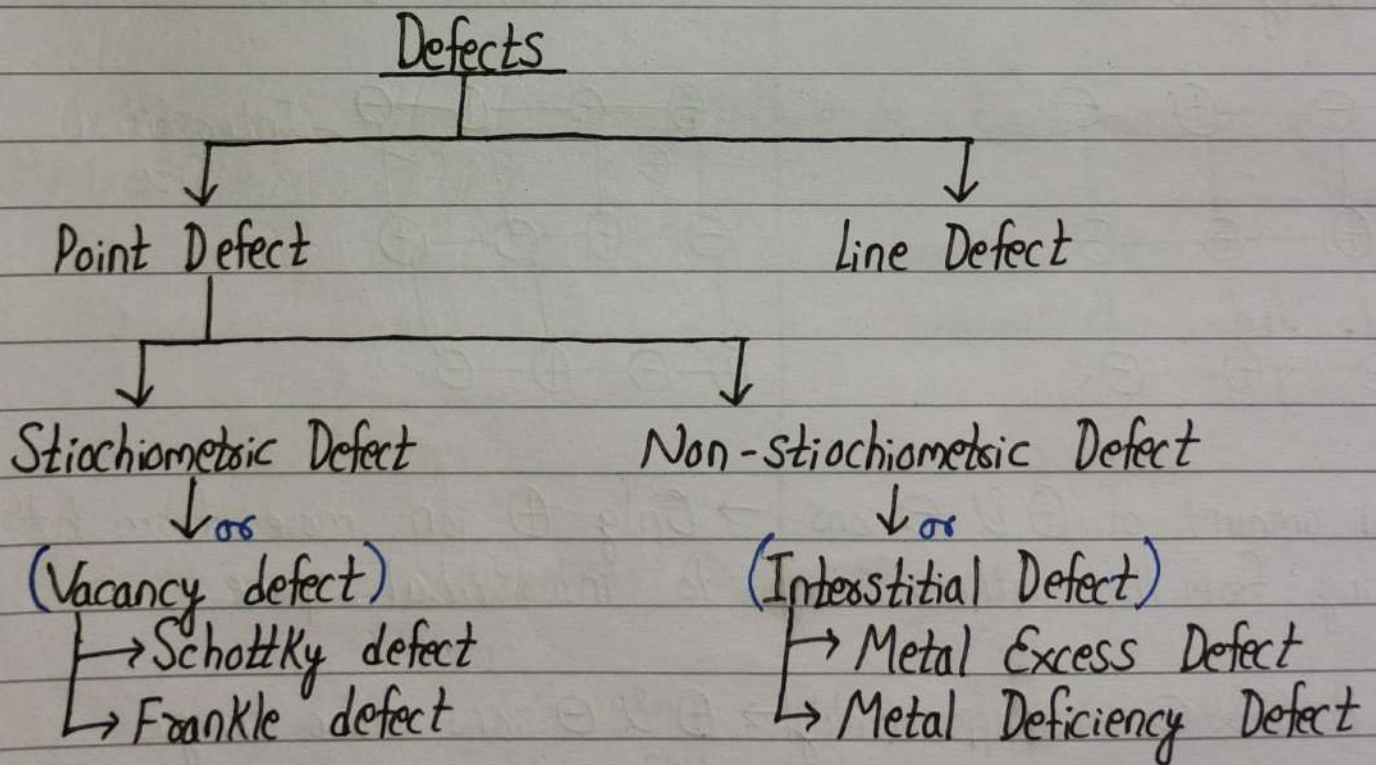
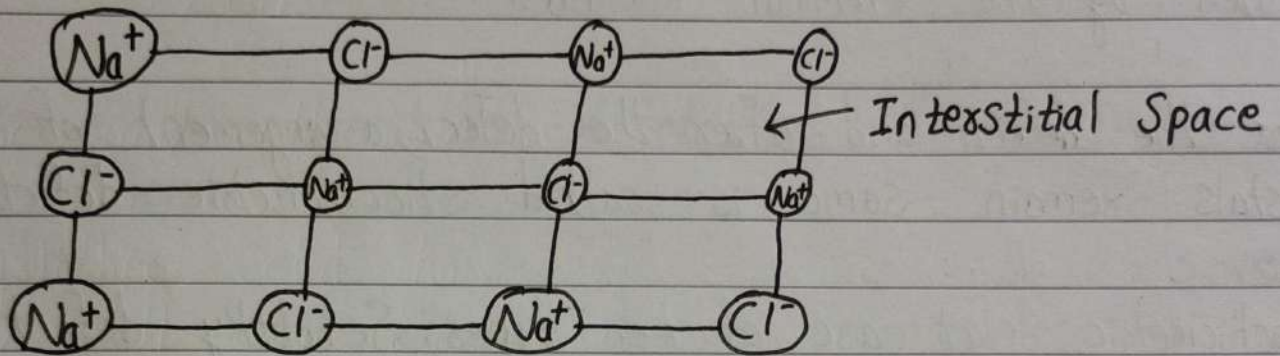
$$\therefore M = \frac{S \times N_A \times a^3}{Z}$$

$$= \frac{7.2 \times 6.022 \times 10^{23} \times (288 \times 10^{-10})^3}{2}$$

$$= 51.78 \text{ g/mol.}$$

Defects in Crystal (Imperfection)

By adding external substances, in a pure crystal to create defect, this is called doping.



Point Defect

Imp

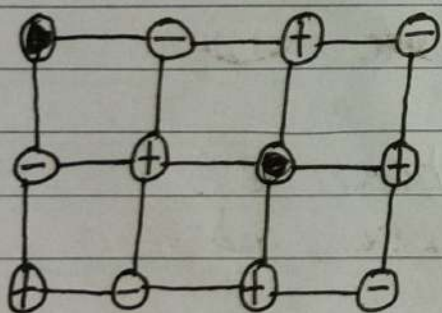
Stoichiometric Defect:

Stoichiometric compounds are those in which the number of positive and negative ions are exactly in ratio indicated by the chemical formula.

Before the defect and after the defect, arrangement of the crystals remain same is called stoichiometric defect.

Stoichiometric defect are of two types; Schottky defect and Frenkel defect.

Schottky Defect



→ Equal amount of $\oplus$ & $\ominus$ ions missing from the lattice.

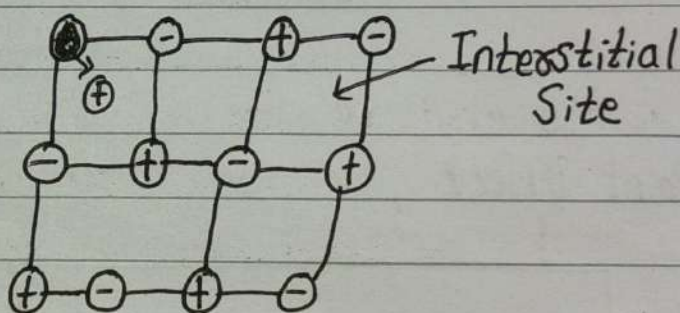
→ Size of $\oplus$ & $\ominus$ ions approximately same.

→ Coordination No. = High

→ Density = Low

→ Examples; NaCl, KCl, CsCl, AgBr

Frenkel Defect



→ Only $\oplus$ ion move from lattice to interstitial space.

→ $\oplus$ & $\ominus$ ions have large difference.

→ Coordination No. = Low

→ Density = Unchanged

→ Examples; AgCl, AgI, ZnS, AgBr

Non-Stoichiometric Defect:

The compounds in which, the ratio of positive and negative ions present in the compound differs from that required by ideal chemical formula of the compound are non-stoichiometric compounds and these defects are called non-stoichiometric defect.

There are two types of defect; Metal Excess Defect and Metal Deficiency Defect.

① Metal Excess Defect

- (i) Due to anionic vacancies
- (ii) Extra cation occupying interstitial site

② Metal Deficiency Defect

- (i) Due to cationic vacancies
- (ii) Extra anion occupying interstitial site

① Metal Excess Defect

(i) due to anionic vacancies (F-centre):

In a metallic crystal for example, NaCl , the metal vapours pass on the surface of the crystal and get ionized. Opposite charge i.e. Cl^- attracted from the crystal to the Na^+ on the surface and now Cl^- ~~due to~~^{by} the diffusion comes on the surface of crystal and create anionic vacancy in the crystal. Now in this condition the $\oplus$ & $-$ ve ratios

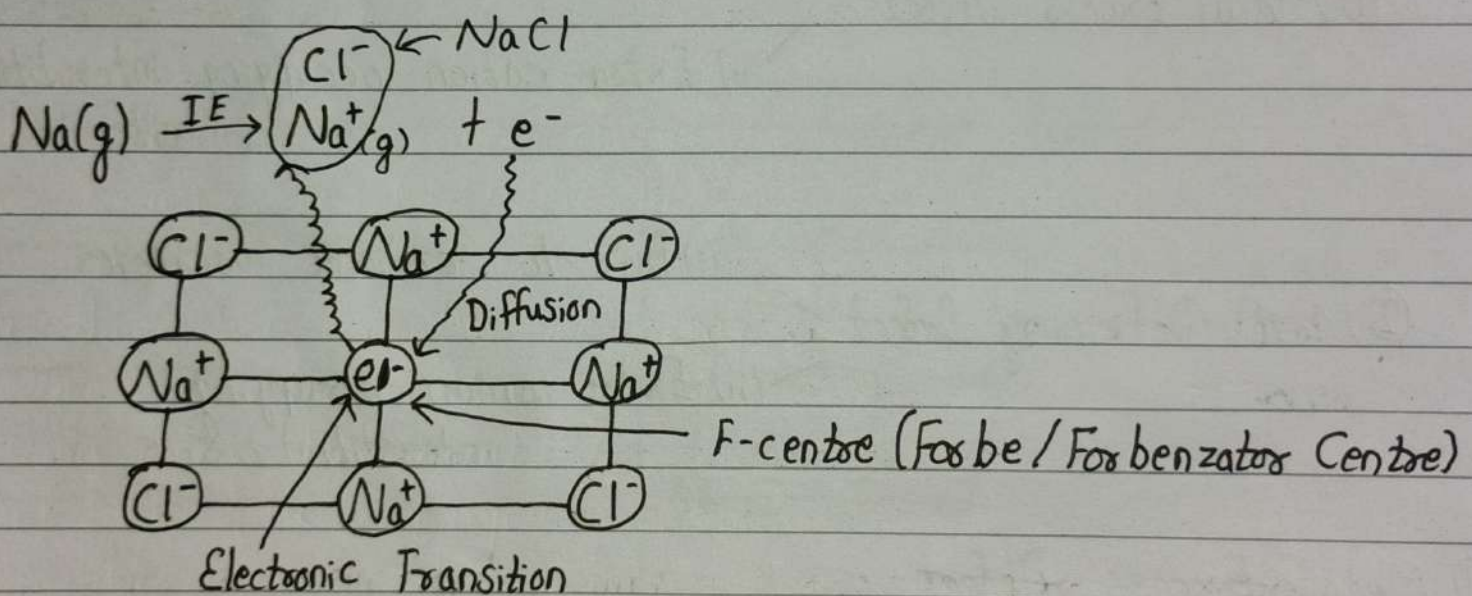
of the crystal become imbalance, electron introduced into the crystal at anionic site and causes electronic transition and impart color. This centre is called F-centre (Förbe/Förbenzator Centre).

NaCl - Yellow

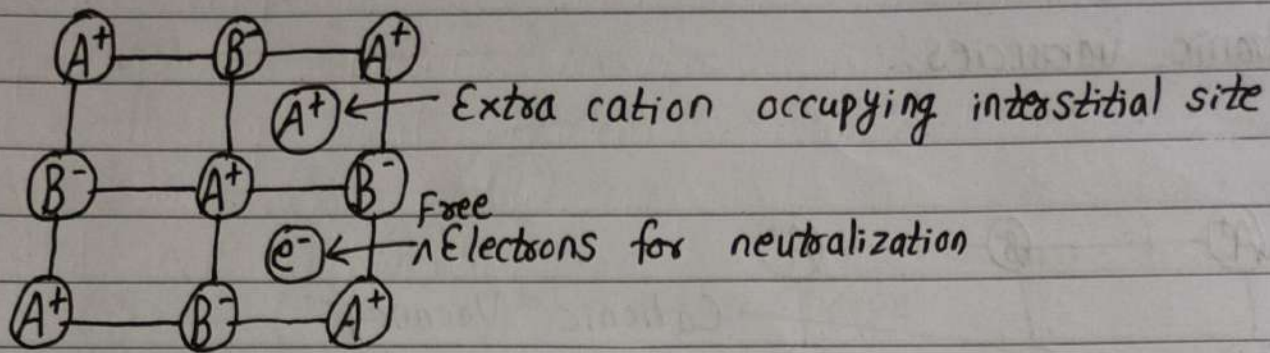
KCl - Violet

LiCl - Pink

This type of defect is observed in those crystal which are likely to form Schottky defect.

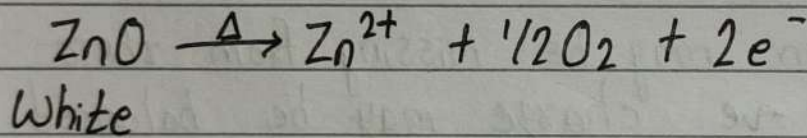


(ii) ~~due~~ Extra cation occupying interstitial site:



In this case, the extra ~~the~~ ion occupies interstitial site and the electrons also occupying another interstitial site to maintain electrical neutrality.

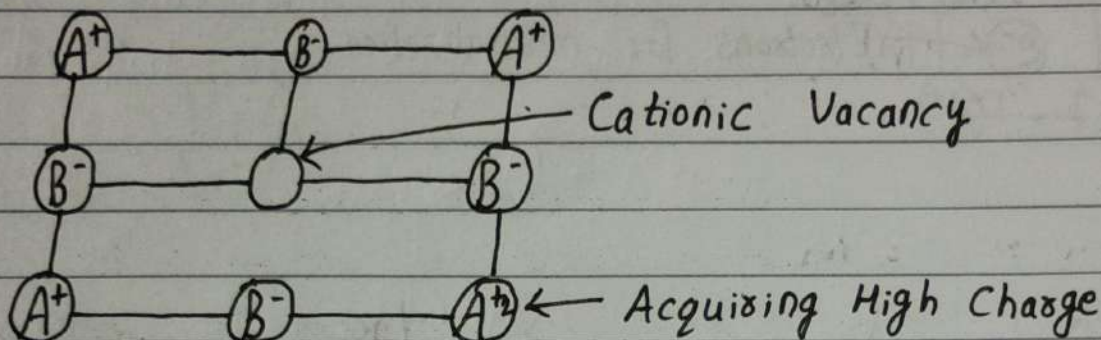
For example;



Zinc oxide heated to give yellow colour because of free electron available in the crystal.

② Metal Deficiency Defect (p-type)

(i) Cationic vacancies:

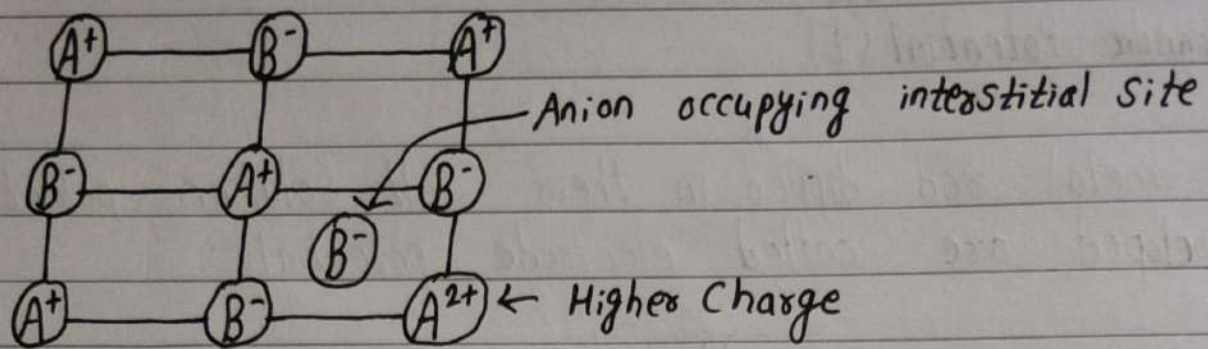


This contains less no. of positive ions than negative ions.
For example; FeS (Fool's Gold), NiO .

In this case, the Fe ion may be missing from their lattice site. The extra -ve charge may be balanced by some near-by metal ion acquiring higher charge i.e. two Fe charges instead of one. This defect is possible in metals which show variable oxidation state.

For example; FeS , NiO .

Extra
(ii) Anion occupying interstitial site:



In this case, the extra anion may be occupying interstitial site. The extra -ve charge is balanced by ~~asce~~ adjacent metal ion which acquire higher charge.